

Unit 3, Lesson 5: Adding and Subtracting Numbers Expressed in Scientific Notation

Benchmarks

MA.B.NSO.1.5 Add, subtract, multiply, and divide numbers expressed in scientific notation with procedural fluency.

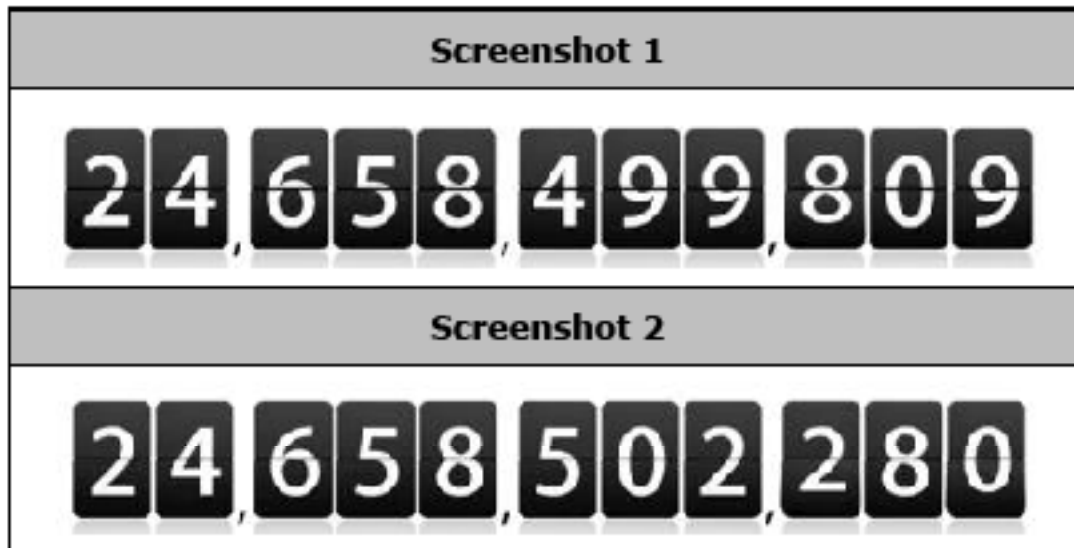
MA.B.NSO.1.6 Solve real-world problems involving operations with numbers expressed in scientific notation.

Learning Targets

- ☐ I can add and subtract numbers expressed in scientific notation.
- ☐ I can solve real-world problems involving operations with numbers expressed in scientific notation.

3.5.1 Warm-Up: 25 Billion Downloads!

1. When the Apple App Store was close to hitting 25 billion downloads, the company posted a counter on its website. As the countdown approached, Daniel took the screenshots shown. Screenshot 2 was taken 4 seconds after screenshot 1.



- a. What do you notice?
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- b. What do you wonder?

3.5.2 Exploration: Measuring the Planets

Work with your partner to answer the following questions.

Diego, Kiran, and Clare were discussing the question shown.

"If Neptune and Saturn were side by side, would they be wider than Jupiter?"

They determined that to answer the question they should add the diameters 4.7×10^4 kilometers (km) and 1.2×10^5 km.

1. Review each student's response, then complete the table.

Student Response	Agree?	Explanation
Diego says, "When we add the distances, we will get $4.7 + 1.2 = 5.9$. The exponent will be 9. So, the two planets are 5.9×10^9 km side by side."	☐ Yes ☐ No	
Kiran wrote 4.7×10^4 as 47,000 and 1.2×10^5 as 120,000 and added them.	☐ Yes ☐ No	
Clare says, "I don't think you can add unless they are the same power of 10." She adds 4.7×10^4 km and 12×10^4 to get 16.7×10^4 .	☐ Yes ☐ No	

2. Jupiter has a diameter of 1.43×10^5 . Which is wider: Neptune and Saturn placed side by side, or Jupiter?

3.5.3 Guided Instruction: Adding and Subtracting Numbers in Scientific Notation

1. Gizmo Gadgets produced 1,500,000 widgets in 2017. In 2018 they produced another 2,000,000.
 - a. How many widgets did they produce during the two-year period?
 - b. Write the answer to Part A using scientific notation.

Scientific notation can be used to add and subtract large and small numbers.

- c. Write the numbers in scientific notation.

$$1,500,000 \qquad \qquad \qquad \underline{\hspace{1cm}} \times 10$$

$$2,000,000 \qquad \qquad \qquad \underline{\hspace{1cm}} \times 10$$

Another way to find the sum of gadgets is to add the numbers using scientific notation. In order to add numbers using scientific notation, the powers of ten should be the same.

- d. Discuss with your partner why, when the powers of ten are not the same, the place value is not the same. If necessary, refer back to the Exploration.
- e. Determine the sum.
 $(1.5 \times 10^6) + (2.0 \times 10^6) = \underline{\hspace{1cm}} \times 10$
- f. Compare your answer to Part B with your answer to Part E. If they are not the same, make any necessary adjustments to your answer.

2. Write each in scientific notation.

Number	Scientific Notation
15,740,000,000	
157.4×10^4	
15.74×10^9	
15.74×10^5	

3. Which numbers in problem 2 are represented by the same scientific notation?

4. Find the sum by filling in the blanks.

$$(6.11 \times 10^5) + (1.574 \times 10^6)$$

$$(6.11 \times 10^5) + (\text{ } \times 10^5)$$

$$= (21.85 \times 10\text{—})$$

$$= (\text{ } \times 10^6)$$

5. The population of New City was 1,625,000 in 2017. The population grew to 2,750,000 by the end of 2018.

a. How many more people lived in New City at the end of 2018 compared to 2017? Write your answer in scientific notation.

b. Write the population numbers from the word problem in scientific notation.

1,625,000 _____ $\times 10$

2,750,000 _____ $\times 10$

c. What is the difference between the populations? Write your answer using scientific notation.

6. Clare was wondering if adding and subtracting numbers in scientific notation would hold true for small numbers as well. The measurements in the table are from her science textbook.

Item	Measurement (centimeters)
Width of human hair strand	0.00108
Diameter of a grain of sand	0.00609
Length of a dust mite	0.0136

- a. Work with your partner to check Clare's work by writing each number in standard form, then confirm the solution.

$$(1.08 \times 10^{-3}) + (6.09 \times 10^{-3}) = (7.17 \times 10^{-3})$$

- b. Find the difference between the length of a dust mite and the width of a human hair strand.

3.5.4 Guided Practice: Adding and Subtracting Numbers Written in Scientific Notation

1. Fill in the blanks to find the difference.

$$(3.8 \times 10^8) - (8.1 \times 10^7)$$

$$(\text{ ______ } \times 10^7) - (8.1 \times 10^7)$$

$$\text{ ______ } \times 10^7$$

$$\text{ ______ } \times 10^8$$

2. Find the difference. Write your answer using scientific notation.

$$1.574 \times 10^{11} - 32.2 \times 10^9$$

3. Find the sum. Write your answer using scientific notation.

$$2.37 \times 10^{22} + 14.29 \times 10^{34}$$

3.5.5 Try It: Adding and Subtracting Numbers Written in Scientific Notation

Add or subtract. Write your answers in scientific notation.

1. $5.3 \times 10^4 + 4.7 \times 10^4$

2. $3.7 \times 10^6 - 3.3 \times 10^6$

3. $4.8 \times 10^{-2} + 6.3 \times 10^{-2}$

4. $6.6 \times 10^{-5} - 6.1 \times 10^{-5}$

5. The table lists the masses, in kilograms (kg), of different creatures on land and in water. Use this information to answer the questions.

Creature	Mass of One Individual (kg)
Ant	3×10^{-6}
Antarctic Krill	4.86×10^{-4}
Zooplankton	5×10^{-8}

- a. What is the difference, in kg, between the mass of an ant and the mass of an Antarctic krill?
- b. What is the total mass of an ant and a zooplankton?

3.5.6 Wrap-Up: Reflection

1. What was the ***muddiest*** (most difficult or confusing) point in today's lesson on adding and subtracting numbers expressed in scientific notation?

